

(28)

$$(i) (AB)^t = B^t A^t$$

Take L.H.S

$$(AB)^t$$

Find we find AB

$$\Rightarrow AB = \begin{bmatrix} 3 & 2 & 1 \\ 4 & -2 & 5 \\ -1 & 0 & -7 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 7 & 4 & 6 \\ 1 & -8 & 5 \\ -1 & 0 & 3 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow AB = \begin{bmatrix} (1)(7) + (2)(1) + (1)(-1) & (1)(4) + (2)(-8) + (1)(0) & (1)(6) + (2)(5) + (1)(3) \\ (4)(7) + (-2)(1) + (0)(-1) & (4)(4) + (-2)(-8) + (0)(0) & (4)(6) + (-2)(5) + (0)(3) \\ (1)(7) + (0)(1) + (-7)(-1) & (1)(4) + (0)(-8) + (-7)(0) & (1)(6) + (0)(5) + (-7)(3) \end{bmatrix}$$

$$\Rightarrow AB = \begin{bmatrix} 37+14-1 & 12-16+0 & 18+10+3 \\ 28-14-5 & 16+16+0 & 24-10+15 \\ 7+0-7 & 4+0-0 & 6+0-21 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow AB = \begin{bmatrix} 40 & -4 & 31 \\ 17 & 32 & 29 \\ -10 & -4 & -21 \end{bmatrix}_{3 \times 3}$$

$$(AB)^t$$

$$= \begin{bmatrix} 40 & 17 & -20 \\ -4 & 32 & -4 \\ 31 & 29 & -21 \end{bmatrix}_{3 \times 3}$$

Take R.H.S

$$A^t = \begin{bmatrix} 7 & 4 & -1 \\ 1 & -8 & 0 \\ 1 & 0 & 3 \end{bmatrix}_{3 \times 3}$$

$$B^t = \begin{bmatrix} 3 & 2 & 1 \\ 4 & -2 & 5 \\ -1 & 0 & -7 \end{bmatrix}_{3 \times 3}$$

Your Success is our win

(29)

$$B^t A^t$$

$$= \begin{bmatrix} 7 & 4 & -1 \\ 1 & -8 & 0 \\ 1 & 0 & 3 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 3 & 2 & 1 \\ 4 & -2 & 5 \\ -1 & 0 & -7 \end{bmatrix}_{3 \times 3}$$

$$= \begin{bmatrix} (7)(3) + (4)(2) + (-1)(-1) & (7)(4) + (4)(-2) + (-1)(0) & (7)(1) + (4)(0) + (-1)(-7) \\ (1)(3) + (-8)(2) + (0)(-1) & (1)(4) + (-8)(-2) + (0)(0) & (1)(1) + (-8)(0) + (0)(-7) \\ (1)(3) + (0)(2) + (3)(-1) & (1)(4) + (0)(-2) + (3)(0) & (1)(1) + (0)(0) + (3)(-7) \end{bmatrix}$$

$$= \begin{bmatrix} 27+14-1 & 28-14-15 & 7+0+7 \\ 3-16 & 4+16+0 & 1-0-0 \\ 3+0-3 & 4-0+0 & 1+0-21 \end{bmatrix}_{3 \times 3}$$

$$= \begin{bmatrix} 40 & 27 & -2 \\ -4 & 20 & 1 \\ 0 & 4 & -20 \end{bmatrix}_{3 \times 3}$$

Ex # 2.3

Evaluate the following determinant.

$$(i) \begin{vmatrix} 5 & 4 & 3 \\ 3 & -4 & 0 \\ 2 & 3 & 1 \end{vmatrix}$$

$$\begin{vmatrix} 5 & 4 & 3 \\ 3 & -4 & 0 \\ 2 & 3 & 1 \end{vmatrix}$$

$$\text{let } |D| = \begin{vmatrix} 5 & 4 & 3 \\ 3 & -4 & 0 \\ 2 & 3 & 1 \end{vmatrix}$$

$$|D| = 5 \begin{vmatrix} -4 & 0 \\ 3 & 1 \end{vmatrix} - 4 \begin{vmatrix} 3 & 0 \\ 2 & 1 \end{vmatrix} + 3 \begin{vmatrix} 3 & -4 \\ 2 & 3 \end{vmatrix}$$

$$|D| = 5[(-4)(1) - (0)(3)] - 4[(3)(1) - (0)(2)] + 3[(3)(3) - (-4)(2)]$$

$$|D| = 5(-4-0) - 4(3-0) + 3(9+8)$$

(30)

$$|D| = -20 - 12 + 51$$

$$|D| = 19 \quad \text{df}$$

$$(ii) \begin{vmatrix} -2 & 4 & 3 \\ 5 & 4 & -2 \\ 2 & 7 & 3 \end{vmatrix}$$

S1

$$\text{Let, } |D| = \begin{vmatrix} -2 & 4 & 3 \\ 5 & 4 & -2 \\ 2 & 7 & 3 \end{vmatrix}$$

$$|D| = -2 \begin{vmatrix} 4 & -2 \\ 7 & 3 \end{vmatrix} - 2 \begin{vmatrix} 5 & -2 \\ 2 & 3 \end{vmatrix} + 3 \begin{vmatrix} 5 & 4 \\ 2 & 7 \end{vmatrix}$$

$$|D| = -2 [(4)(3) - (-2)(7)] - 2 [(5)(3) - (2)(-2)] + 3 [(5)(7) - (2)(4)]$$

$$|D| = -2 (12 + 14) - 2 (15 + 4) + 3 (35 - 8)$$

$$|D| = -52 - 38 + 81$$

$$|D| = -47 \quad \text{df}$$

$$(iii) \begin{vmatrix} 2a & c & c \\ a & a & 2a \\ b & 2b & b \end{vmatrix}$$

S1

$$\text{Let, } |D| = \begin{vmatrix} 2a & c & c \\ a & a & 2a \\ b & 2b & b \end{vmatrix}$$

$$|D| = 2a \begin{vmatrix} a & 2a \\ 2b & b \end{vmatrix} - c \begin{vmatrix} a & 2a \\ b & b \end{vmatrix} + c \begin{vmatrix} a & a \\ b & 2b \end{vmatrix}$$

$$|D| = 2a [(a)(b) - (2a)(2b)] - c [(a)(b) - (2a)(b)] + c [(a)(2b) - (b)(a)]$$

$$|D| = 2a (ab - 4ab) - c (ab - 2ab) + c (2ab - ab)$$

$$|D| = -6abc + abc + abc$$

$$|D| = -4abc$$

(31)

$$|D| = -6abc + 2abc$$

$$|D| = -4abc \quad \text{df}$$

$$(ii) \begin{vmatrix} 1 & -i & 1-i \\ 0 & 1 & i \\ 1+i & i & 1 \end{vmatrix}$$

S1

$$\text{Let, } |D| = \begin{vmatrix} 1 & -i & 1-i \\ 0 & 1 & i \\ 1+i & i & 1 \end{vmatrix}$$

$$|D| = 1 \begin{vmatrix} 1 & i \\ i & 1 \end{vmatrix} - i \begin{vmatrix} 0 & 1 \\ 1+i & 1 \end{vmatrix} + (1-i) \begin{vmatrix} 0 & 1 \\ 1+i & i \end{vmatrix}$$

$$|D| = 1 [(1)(1) - (i)(i)] - i [0(1) - (1)(1+i)] + (1-i) [(0)(i) - (1)(1+i)]$$

$$|D| = 1 (1 + 1) - i (0 - 1 + 1) + (1-i) (0 - 1 - i)$$

$$|D| = 1 + 1 - i + i + i^2 + 2 - 1 - i + i^2 + i^2$$

$$|D| = 2 - 1 - 1$$

$$|D| = 0 \quad \text{df}$$

$$(iv) \begin{vmatrix} 7 & -2 & 1 \\ 2 & 2 & 4 \\ 4 & 3 & 7 \end{vmatrix}$$

S1

$$\text{Let, } |D| = \begin{vmatrix} 7 & -2 & 1 \\ 2 & 2 & 4 \\ 4 & 3 & 7 \end{vmatrix}$$

$$|D| = 7 \begin{vmatrix} 2 & 4 \\ 3 & 7 \end{vmatrix} - 2 \begin{vmatrix} 2 & 4 \\ 4 & 7 \end{vmatrix} + 1 \begin{vmatrix} 2 & 2 \\ 4 & 3 \end{vmatrix}$$

$$|D| = 7 [(2)(7) - (3)(4)] - 2 [(2)(7) - (4)(4)] + 1 [(2)(3) - (4)(2)]$$

$$|D| = 7 (14 - 12) - 2 (14 - 16) + 1 (6 - 8)$$

(30)

$$|D| = -20 - 12 + 51$$

$$|D| = 19 \quad \text{df}$$

$$(ii) \begin{vmatrix} -2 & 4 & 3 \\ 5 & 4 & -2 \\ 2 & 7 & 3 \end{vmatrix}$$

S1

$$\text{Let, } |D| = \begin{vmatrix} -2 & 4 & 3 \\ 5 & 4 & -2 \\ 2 & 7 & 3 \end{vmatrix}$$

$$|D| = -2 \begin{vmatrix} 4 & -2 \\ 7 & 3 \end{vmatrix} - 2 \begin{vmatrix} 5 & -2 \\ 2 & 3 \end{vmatrix} + 3 \begin{vmatrix} 5 & 4 \\ 2 & 7 \end{vmatrix}$$

$$|D| = -2 [(4)(3) - (-2)(7)] - 2 [(5)(3) - (-2)(2)] + 3 [(5)(7) - (2)(4)]$$

$$|D| = -2 (12 + 14) - 2 (15 + 4) + 3 (35 - 8)$$

$$|D| = -52 - 38 + 81$$

$$|D| = -47 \quad \text{df}$$

$$(iii) \begin{vmatrix} 2a & c & c \\ a & a & 2a \\ b & 2b & b \end{vmatrix}$$

S1

$$\text{Let, } |D| = \begin{vmatrix} 2a & c & c \\ a & a & 2a \\ b & 2b & b \end{vmatrix}$$

$$|D| = 2a \begin{vmatrix} a & 2a \\ 2b & b \end{vmatrix} - c \begin{vmatrix} a & 2a \\ b & b \end{vmatrix} + c \begin{vmatrix} a & a \\ b & 2b \end{vmatrix}$$

$$|D| = 2a [(a)(b) - (2a)(2b)] - c [(a)(b) - (2a)(b)] + c [(a)(2b) - (b)(a)]$$

$$|D| = 2a (ab - 4ab) - c (ab - 2ab) + c (2ab - ab)$$

$$|D| = -6abc + abc + abc$$

$$|D| = -4abc$$

(31)

$$|D| = -6abc + 2abc$$

$$|D| = -4abc \quad \text{df}$$

$$(ii) \begin{vmatrix} 1 & -i & 1-i \\ 0 & 1 & i \\ 1+i & i & 1 \end{vmatrix}$$

S1

$$\text{Let, } |D| = \begin{vmatrix} 1 & -i & 1-i \\ 0 & 1 & i \\ 1+i & i & 1 \end{vmatrix}$$

$$|D| = 1 \begin{vmatrix} 1 & i \\ i & 1 \end{vmatrix} - i \begin{vmatrix} 0 & 1 \\ 1+i & 1 \end{vmatrix} + (1-i) \begin{vmatrix} 0 & 1 \\ 1+i & i \end{vmatrix}$$

$$|D| = 1 [(1)(1) - (i)(i)] - i [0(1) - (1)(1+i)] + (1-i) [(0)(i) - (1)(1+i)]$$

$$|D| = 1 (1 + 1) - i (0 - 1 + 1) + (1-i) (0 - 1 - i)$$

$$|D| = 1 + 1 - i + i + i^2 + 2 - 1 - i + i^2 + i^2$$

$$|D| = 2 - 1 - 1$$

$$|D| = 0 \quad \text{df}$$

$$(iv) \begin{vmatrix} 7 & -2 & 1 \\ 2 & 2 & 4 \\ 4 & 3 & 7 \end{vmatrix}$$

S1

$$\text{Let, } |D| = \begin{vmatrix} 7 & -2 & 1 \\ 2 & 2 & 4 \\ 4 & 3 & 7 \end{vmatrix}$$

$$|D| = 7 \begin{vmatrix} 2 & 4 \\ 3 & 7 \end{vmatrix} - 2 \begin{vmatrix} 2 & 4 \\ 4 & 7 \end{vmatrix} + 1 \begin{vmatrix} 2 & 2 \\ 4 & 3 \end{vmatrix}$$

$$|D| = 7 [(2)(7) - (3)(4)] - 2 [(2)(7) - (4)(4)] + 1 [(2)(3) - (4)(2)]$$

$$|D| = 7 (14 - 12) - 2 (14 - 16) + 1 (6 - 8)$$

32

$$|D| = 14 - 4 - 2$$

$$|D| = 8 \text{ pf}$$

$$(vi) \begin{vmatrix} 1 & 1 & 1 \\ w & w^2 & 1 \\ w^2 & w & 1 \end{vmatrix}$$

$$\text{S.1} \\ \text{let, } |D| = \begin{vmatrix} 1 & 1 & 1 \\ w & w^2 & 1 \\ w^2 & w & 1 \end{vmatrix}$$

$$|D| = 1 \begin{vmatrix} w^2 & 1 \\ w & 1 \end{vmatrix} - 1 \begin{vmatrix} w & 1 \\ w^2 & 1 \end{vmatrix} + 1 \begin{vmatrix} w & w^2 \\ w^2 & w \end{vmatrix}$$

$$|D| = 1[(w^2)(1) - (w)(1)] - 1[(w)(1) - (w^2)(1)] + 1[(w)(w) - (w^2)(w^2)]$$

$$|D| = 1(w^2 - w) - 1(w - w^2) + 1(w^2 - w^4) \quad \because w^4 = w$$

$$|D| = w^2 - w - w + w^2 + w^2 - w$$

$$|D| = 3w^2 - 3w$$

$$|D| = 3(w^2 - w) \text{ pf}$$

② Identify the singular or non-singular matrix.

$$(i) \begin{bmatrix} 4 & 0 & 1 \\ 7 & 5 & 5 \\ -12 & -6 & -7 \end{bmatrix}$$

$$\text{let } A = \begin{bmatrix} 4 & 0 & 1 \\ 7 & 5 & 5 \\ -12 & -6 & -7 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 4 & 0 & 1 \\ 7 & 5 & 5 \\ -12 & -6 & -7 \end{vmatrix}$$

$$|A| = 4 \begin{vmatrix} 5 & 5 \\ -6 & -7 \end{vmatrix} - 0 \begin{vmatrix} 7 & 5 \\ -12 & -7 \end{vmatrix} + 1 \begin{vmatrix} 7 & 5 \\ -12 & -6 \end{vmatrix}$$

$$|A| = 4[(5)(-7) - (5)(-6)] - 0[(7)(-7) - (-12)(5)] + 1[(7)(-6) - (-6)(-12)]$$

33

$$|A| = 4(-35 + 30) - 0(-47 + 60) + 1(42 + 60)$$

$$|A| = 4(-5) - 0 + 1(102)$$

$$|A| = -20 + 102$$

$$|A| = -2 \text{ Non-singular matrix.}$$

$$(ii) \begin{bmatrix} 4 & 2 & 0 \\ 3 & 0 & 1 \\ -1 & 1 & -1 \end{bmatrix}$$

$$\text{S.1} \\ \text{let, } A = \begin{bmatrix} 4 & 2 & 0 \\ 3 & 0 & 1 \\ -1 & 1 & -1 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 4 & 2 & 0 \\ 3 & 0 & 1 \\ -1 & 1 & -1 \end{vmatrix}$$

$$|A| = 4 \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} - 2 \begin{vmatrix} 3 & 1 \\ -1 & -1 \end{vmatrix} + 0 \begin{vmatrix} 3 & 0 \\ -1 & 1 \end{vmatrix}$$

$$|A| = 4[(0)(-1) - (1)(1)] - 2[(3)(-1) - (-1)(-1)] + 0[(3)(1) - (-1)(0)]$$

$$|A| = 4(-1) - 2(-3 + 1) + 0(3 + 1)$$

$$|A| = -4 + 4$$

$$|A| = 0, \text{ As } |A| = 0 \text{ So, } A \text{ is a singular matrix.}$$

$$(iii) \begin{bmatrix} 13 & -5 & 4 \\ 8 & 1 & 3 \\ 7 & -1 & 2 \end{bmatrix}$$

$$\text{S.1} \\ \text{let } A = \begin{bmatrix} 13 & -5 & 4 \\ 8 & 1 & 3 \\ 7 & -1 & 2 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 13 & -5 & 4 \\ 8 & 1 & 3 \\ 7 & -1 & 2 \end{vmatrix}$$

$$|A| = 13 \begin{vmatrix} 1 & 3 \\ -1 & 2 \end{vmatrix} + 5 \begin{vmatrix} 8 & 3 \\ 7 & 2 \end{vmatrix} + 4 \begin{vmatrix} 8 & 1 \\ 7 & -1 \end{vmatrix}$$

$$|A| = 13[(1)(2) - (-1)(3)] + 5[(8)(2) - (7)(3)] + 4[(-8)(-1) - (7)(1)]$$

34

$$|A| = 13(2+5) + 5(16-21) + 4(-8-7)$$

$$|A| = 13(7) + 5(-5) + 4(-15)$$

$$|A| = 91 - 25 - 60$$

$|A| = -20$, As $|A| \neq 0$ So, A is non-singular.

$$(iv) \begin{bmatrix} 1 & 6 & 3 \\ -2 & 1 & 0 \\ 6 & 4 & 2 \end{bmatrix}$$

Sol

$$\text{let } A = \begin{bmatrix} 1 & 6 & 3 \\ -2 & 1 & 0 \\ 6 & 4 & 2 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & 6 & 3 \\ -2 & 1 & 0 \\ 6 & 4 & 2 \end{vmatrix}$$

$$|A| = 1 \begin{vmatrix} 1 & 0 & -2 \\ 4 & 2 & 6 \end{vmatrix} - 6 \begin{vmatrix} -2 & 0 \\ 6 & 2 \end{vmatrix} + 3 \begin{vmatrix} -2 & 1 \\ 6 & 4 \end{vmatrix}$$

$$|A| = 1[(1)(2) - (-4)(6)] - 6[(-2)(2) - (6)(6)] + 3[(-2)(4) - (6)(1)]$$

$$|A| = 1(2+24) - 6(-4-36) + 3(-8-6)$$

$$|A| = 2 + 24 - 42$$

$|A| = -16$ As $|A| \neq 0$, So, A is a non-singular.

$$(v) \begin{bmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Sol

$$\text{let } A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{vmatrix}$$

$$|A| = 4 \begin{vmatrix} 3 & 0 \\ 0 & 2 \end{vmatrix} - 0 \begin{vmatrix} 0 & 0 \\ 0 & 2 \end{vmatrix} + 0 \begin{vmatrix} 0 & 3 \\ 0 & 0 \end{vmatrix}$$

$$|A| = 4(6-0) - 0(0-0) + 0(0-0)$$

$$|A| = 24$$

So, A is singular.

35

$$(vi) \begin{bmatrix} 20 & 10 & 20 \\ 2 & 1 & 3 \\ 0 & 20 & 1 \end{bmatrix}$$

Sol

$$\text{let } A = \begin{bmatrix} 20 & 10 & 20 \\ 2 & 1 & 3 \\ 0 & 20 & 1 \end{bmatrix}$$

Now,

$$|A| = \begin{vmatrix} 20 & 10 & 20 \\ 2 & 1 & 3 \\ 0 & 20 & 1 \end{vmatrix}$$

$$|A| = 20 \begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix} - 10 \begin{vmatrix} 2 & 3 \\ 0 & 1 \end{vmatrix} + 20 \begin{vmatrix} 2 & 1 \\ 0 & 20 \end{vmatrix}$$

$$|A| = 20[(1)(1) - (3)(0)] - 10[(2)(1) - (3)(0)] + 20[(2)(20) - (1)(0)]$$

$$|A| = 20(1-0) - 10(2-0) + 20(40-0)$$

$$|A| = 20 - 20 + 800$$

$|A| = 800$, As $|A| \neq 0$, So A is non-singular.

(3) Find the value of n for which following matrix are singular.

$$(i) \begin{bmatrix} 5 & 4 \\ x & 8 \end{bmatrix}$$

Sol

$$\text{let } A = \begin{bmatrix} 5 & 4 \\ x & 8 \end{bmatrix}$$

$$\therefore |A| = 0$$

$$\begin{vmatrix} 5 & 4 \\ x & 8 \end{vmatrix} = 0$$

$$\Rightarrow 40 - 4x = 0$$

$$\Rightarrow 40 = 4x$$

$$\Rightarrow \frac{40}{4} = x$$

$$\Rightarrow \boxed{x = 10}$$

$$(ii) \begin{bmatrix} 10 & 5 \\ 6 & x \end{bmatrix}$$

Sol

$$\text{let } A = \begin{bmatrix} 10 & 5 \\ 6 & x \end{bmatrix}$$

$$\therefore |A| = 0$$

$$\begin{vmatrix} 10 & 5 \\ 6 & x \end{vmatrix} = 0$$

$$\Rightarrow 10x - 30 = 0$$

$$\Rightarrow x = 3$$

$$\Rightarrow \boxed{x = 3}$$

36

$$(iii) \begin{vmatrix} 6 & 3 & 7 \\ 2 & -4 & 2 \\ 5 & x & 1 \end{vmatrix}$$

Sol

$$\text{let } A = \begin{bmatrix} 6 & 3 & 7 \\ 2 & -4 & 2 \\ 5 & x & 1 \end{bmatrix}_{3 \times 3}$$

$$\therefore |A| = 0$$

$$\Rightarrow \begin{vmatrix} 6 & 3 & 7 \\ 2 & -4 & 2 \\ 5 & x & 1 \end{vmatrix} = 0$$

$$\Rightarrow 6 \begin{vmatrix} -4 & 2 \\ x & 1 \end{vmatrix} - 3 \begin{vmatrix} 2 & 2 \\ 5 & 1 \end{vmatrix} + 7 \begin{vmatrix} 2 & -4 \\ 5 & x \end{vmatrix} = 0$$

$$\Rightarrow 6[(-4)(1) - (2)(x)] - 3[(2)(1) - (2)(5)] + 7[(2)(x) - (-4)(5)] = 0$$

$$\Rightarrow 6(-4 - 2x) - 3(2 - 10) + 7(2x + 20) = 0$$

$$\Rightarrow -24 - 12x - 9(-8) + 14x + 140 = 0$$

$$\Rightarrow -24 + 72 - 12x + 14x + 140 = 0$$

$$\Rightarrow 188 + 2x = 0$$

$$\Rightarrow x = -94$$

Ans

$$(iv) \begin{vmatrix} x & -2 & 1 \\ 2 & -5 & 4 \\ x & -2 & -1 \end{vmatrix}$$

Sol

$$\text{let } A = \begin{bmatrix} x & -2 & 1 \\ 2 & -5 & 4 \\ x & -2 & -1 \end{bmatrix}_{3 \times 3}$$

$$\therefore |A| = 0$$

$$\begin{vmatrix} x & -2 & 1 \\ 2 & -5 & 4 \\ x & -2 & -1 \end{vmatrix} = 0$$

37

$$\Rightarrow x \begin{vmatrix} -3 & 4 \\ -2 & -1 \end{vmatrix} + 2 \begin{vmatrix} 2 & 4 \\ x & -1 \end{vmatrix} + 1 \begin{vmatrix} 2 & -2 \\ x & -2 \end{vmatrix} = 0$$

$$\Rightarrow x[(-3)(-1) - (-2)(4)] + 2[(2)(-1) - (4)(x)] + 1[(2)(-2) - (-2)(x)] = 0$$

$$\Rightarrow x(3 + 8) + 2(-2 - 4x) + 1(-4 + 2x) = 0$$

$$\Rightarrow 11x - 4 - 8x - 4 + 2x = 0$$

$$\Rightarrow 5x - 8 = 0$$

$$\Rightarrow x = \frac{8}{5}$$

$$\Rightarrow x = 1.6$$

6) Find adjoint of the following matrices.

$$(i) \begin{bmatrix} 3 & 4 \\ -1 & 2 \end{bmatrix}$$

Sol

$$\text{let } A = \begin{bmatrix} 3 & 4 \\ -1 & 2 \end{bmatrix}_{2 \times 2}$$

$$\text{Adj of } A = \begin{bmatrix} 2 & -4 \\ 1 & 3 \end{bmatrix}_{2 \times 2}$$

$$(ii) \begin{bmatrix} 7 & 6 & 8 \\ 1 & 3 & 2 \\ 2 & 7 & 5 \end{bmatrix}$$

Sol

$$\text{let } A = \begin{bmatrix} 7 & 6 & 8 \\ 1 & 3 & 2 \\ 2 & 7 & 5 \end{bmatrix}_{3 \times 3}$$

$$\text{Co-factor of } A$$

$$A_{ij} = (-1)^{i+j} M_{ij}$$

$$(iii) \begin{bmatrix} -0.5 & 0.5 \\ 1 & 2 \end{bmatrix}$$

Sol

$$\text{let } A = \begin{bmatrix} -0.5 & 0.5 \\ 1 & 2 \end{bmatrix}_{2 \times 2}$$

$$\text{Adj of } A = \begin{bmatrix} 2 & -0.5 \\ -1 & -0.5 \end{bmatrix}_{2 \times 2}$$

$$A_{11} = (-1)^{1+1} \begin{vmatrix} 2 & 2 \\ 7 & 5 \end{vmatrix}$$

$$A_{11} = 1$$

$$A_{12} = (-1)^{1+2} \begin{vmatrix} 1 & 1 \\ 2 & 5 \end{vmatrix}$$

(38)

$$A_{12} = (-1)^3 (5-4)$$

$$A_{12} = -1$$

$$A_{13} = (-1)^{1+3} \begin{vmatrix} 1 & 3 \\ 2 & 7 \end{vmatrix}$$

$$A_{13} = (-1)^4 (7-6)$$

$$A_{13} = 1$$

$$A_{21} = (-1)^{2+1} \begin{vmatrix} 6 & 8 \\ 7 & 5 \end{vmatrix}$$

$$A_{21} = (-1)^3 (30-56)$$

$$A_{21} = -26$$

$$A_{22} = (-1)^{2+2} \begin{vmatrix} 4 & 9 \\ 2 & 5 \end{vmatrix}$$

$$A_{22} = (-1)^4 (20-18)$$

$$A_{22} = 2$$

$$A_{23} = (-1)^{2+3} \begin{vmatrix} 4 & 6 \\ 2 & 7 \end{vmatrix}$$

$$A_{23} = (-1)^5 (28-12)$$

$$A_{23} = -16$$

$$A_{11} = (-1)^{1+1} \begin{vmatrix} 6 & 8 \\ 3 & 2 \end{vmatrix}$$

$$A_{21} = (-1)^4 (12-24)$$

$$A_{21} = -12$$

$$A_{32} = (-1)^{3+2} \begin{vmatrix} 4 & 8 \\ 1 & 2 \end{vmatrix}$$

$$A_{33} = (-1)^5 (9-8)$$

$$A_{32} = 0$$

$$A_{33} = (-1)^{3+3} \begin{vmatrix} 4 & 6 \\ 1 & 3 \end{vmatrix}$$

$$A_{33} = (-1)^6 (12-6)$$

$$A_{31} = 6$$

$$\text{Adj. of } A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^t$$

$$\text{Adj. of } A = \begin{bmatrix} 1 & -1 & 1 \\ 26 & 4 & -16 \\ -12 & 0 & 6 \end{bmatrix}^t$$

$$\text{Adj. of } A = \begin{bmatrix} 1 & 26 & -12 \\ -1 & 4 & 0 \\ 1 & -16 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1-i \\ 1+i & 5 & 1 \\ 0 & 1 & -i \end{bmatrix}$$

SOL

$$\text{Let } A = \begin{bmatrix} 1 & 0 & 1-i \\ 1+i & 5 & 1 \\ 0 & 1 & -i \end{bmatrix}$$

Co-factors of A:

$$A_{11} = (-1)^{1+1} \begin{vmatrix} 5 & 1 \\ 1 & -i \end{vmatrix}$$

$$A_{11} = (-1)^3 (1-i)$$

$$A_{11} = 0$$

$$A_{12} = (-1)^{1+2} \begin{vmatrix} 1+i & 1 \\ 0 & -i \end{vmatrix}$$

$$A_{12} = (-1)^3 (-i+1-0)$$

$$A_{12} = -i+1$$

$$A_{13} = (-1)^{1+3} \begin{vmatrix} 1+i & 5 \\ 0 & 1 \end{vmatrix}$$

$$A_{13} = (-1)^4 (1+i-0)$$

$$A_{13} = 1+i$$

$$A_{21} = (-1)^{2+1} \begin{vmatrix} 0 & 1-i \\ 1 & -i \end{vmatrix}$$

$$A_{21} = (-1)^3 (0-1+i)$$

$$A_{21} = 1-i$$

$$A_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 1-i \\ 0 & -i \end{vmatrix}$$

$$A_{22} = (-1)^4 (-i-0)$$

$$A_{22} = -i$$

$$A_{23} = (-1)^{2+3} \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

$$A_{23} = (-1)^5 (1-0)$$

$$A_{23} = -1$$

$$A_{31} = (-1)^{3+1} \begin{vmatrix} 0 & 1-i \\ i & 1 \end{vmatrix}$$

$$A_{31} = (-1)^4 (0-i+1)$$

$$A_{31} = -1-i$$

$$A_{32} = (-1)^{3+2} \begin{vmatrix} 1 & 1-i \\ 1+i & 1 \end{vmatrix}$$

$$A_{32} = (-1)^5 [1-(1+i)(1-i)]$$

$$A_{32} = -1 [1-(1-i+i-i^2)]$$

$$A_{32} = -1 (1-2)$$

$$A_{32} = 1$$

$$A_{33} = (-1)^{3+3} \begin{vmatrix} 1 & 0 \\ 1+i & i \end{vmatrix}$$

$$A_{33} = (-1)^6 (i-0)$$

$$A_{33} = i$$

(40)

$$\text{Adj of } A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^t$$

$$A_{11} = (-1)^{1+1} \begin{vmatrix} 0 & 6 \\ 0 & 0 \end{vmatrix}$$

$$A_{12} = (-1)^{1+2} (0-0)$$

$$A_{13} = 0$$

$$\text{Adj of } A = \begin{bmatrix} 0 & 3-1 & 1+3 \\ 1-3 & -3 & -1 \\ 2-3 & 1 & 3 \end{bmatrix}$$

$$A_{21} = (-1)^{2+1} \begin{vmatrix} 0 & 0 \\ 0 & 7 \end{vmatrix}$$

$$A_{22} = (-1)^{2+2} (0-0)$$

$$A_{23} = 0$$

$$\text{Adj of } A = \begin{bmatrix} 0 & 1-3 & -3-1 \\ 3-1 & -3 & 1 \\ 1-3 & 1 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

$$\text{let } A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

Co-factor of A:

$$A_{ij} = (-1)^{i+j} M_{ij}$$

$$A_{11} = (-1)^{1+1} \begin{vmatrix} 6 & 0 \\ 0 & 7 \end{vmatrix}$$

$$A_{12} = (-1)^{1+2} (42-0)$$

$$A_{13} = 42$$

$$A_{21} = (-1)^{2+1} \begin{vmatrix} 0 & 0 \\ 0 & 7 \end{vmatrix}$$

$$A_{22} = (-1)^{2+2} (0-0)$$

$$A_{23} = 0$$

$$A_{31} = (-1)^{3+1} \begin{vmatrix} 5 & 0 \\ 0 & 0 \end{vmatrix}$$

$$A_{32} = (-1)^{3+2} (0-0)$$

$$A_{33} = 0$$

$$A_{31} = (-1)^{3+1} \begin{vmatrix} 0 & 0 \\ 6 & 0 \end{vmatrix}$$

$$A_{32} = (-1)^{3+2} (0-0)$$

$$A_{33} = 0$$

(41)

$$A_{11} = (-1)^{1+1} \begin{vmatrix} 5 & 0 \\ 0 & 0 \end{vmatrix}$$

$$A_{12} = (-1)^{1+2} (0-0)$$

$$A_{13} = 0$$

$$A_{21} = (-1)^{2+1} \begin{vmatrix} 5 & 0 \\ 0 & 6 \end{vmatrix}$$

$$A_{22} = (-1)^{2+2} (30-0)$$

$$A_{23} = 30$$

$$\text{Adj of } A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^t$$

$$\text{Adj of } A = \begin{bmatrix} 42 & 0 & 0 \\ 0 & 35 & 0 \\ 0 & 0 & 30 \end{bmatrix}$$

$$\text{Adj of } A = \begin{bmatrix} 42 & 0 & 0 \\ 0 & 35 & 0 \\ 0 & 0 & 30 \end{bmatrix}$$

$$\text{If } A = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

then verify

$$(A^{-1})^T = A$$

CA

Take L.H.S

$$A = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 5 & 0 \\ 0 & 5 \end{vmatrix}$$

$$|A| = (5)(5) - (0)(0)$$

$$|A| = 25 - 0$$

$$|A| = 25 \neq 0 \therefore \text{Inverse exists}$$

$$\text{Adj of } A = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \times \text{Adj of } A$$

$$A^{-1} = \frac{1}{25} \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 5 \times \frac{1}{25} & 0 \times \frac{1}{25} \\ 0 \times \frac{1}{25} & 5 \times \frac{1}{25} \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{1}{5} & 0 \\ 0 & \frac{1}{5} \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{1}{5} & 0 \\ 0 & \frac{1}{5} \end{bmatrix}$$

$$|A^{-1}| = \left(\frac{1}{5} \right) \left(\frac{1}{5} \right) - (0)(0)$$

$$|A^{-1}| = \frac{1}{25} - 0$$

$$|A^{-1}| = \frac{1}{25} \neq 0$$

$$|A^{-1}| \neq 0 \therefore \text{Inverse exists}$$

42

$$\text{Adj. of } A^{-1} = \begin{bmatrix} -4/15 & -2/3 \\ -7/15 & -1/3 \end{bmatrix}$$

$$\Rightarrow AB = \begin{bmatrix} 11+20 & 23+10 \\ 7+10 & 21+5 \end{bmatrix}_{2 \times 2}$$

$$\Rightarrow AB = \begin{bmatrix} 31 & 33 \\ 17 & 26 \end{bmatrix}_{2 \times 2}$$

$$(A^{-1})^{-1} = \frac{1}{|A^{-1}|} \times \text{Adj. of } A^{-1}$$

Now, for $(AB)^{-1}$:

$$(A^{-1})^{-1} = \frac{1}{-3/45} \times \begin{bmatrix} -4/15 & -2/3 \\ -7/15 & -1/3 \end{bmatrix}$$

$$|AB| = \begin{vmatrix} 31 & 33 \\ 17 & 26 \end{vmatrix}$$

$$(A^{-1})^{-1} = -45 \times \begin{bmatrix} -4/15 & -2/3 \\ -7/15 & -1/3 \end{bmatrix}$$

$$|AB| = (31)(26) - (17)(33)$$

$$|AB| = 806 - 731$$

$$(A^{-1})^{-1} = \begin{bmatrix} 11/15 \times -45/3 & -2/3 \times -45/3 \\ -7/15 \times -45/3 & -1/3 \times -45/3 \end{bmatrix}$$

$$|AB| = 75, |AB| \neq 0 \text{ so it's inverse exist}$$

$$(A^{-1})^{-1} = \begin{bmatrix} 11 & 10 \\ 7 & 5 \end{bmatrix}$$

$$\text{Adj. of } AB = \begin{bmatrix} 26 & -43 \\ -17 & 31 \end{bmatrix}_{2 \times 2}$$

R.H.S

Hence, proved!

$$(AB)^{-1} = \frac{1}{|AB|} \times \text{Adj. of } |AB|$$

$$(ii) (AB)^{-1} = B^{-1}A^{-1}$$

$$(AB)^{-1} = \frac{1}{75} \begin{bmatrix} 26 & -43 \\ -17 & 31 \end{bmatrix}_{2 \times 2}$$

Take L.H.S

$$(AB)^{-1} = \begin{bmatrix} 26/75 & -43/75 \\ -17/75 & 31/75 \end{bmatrix}_{2 \times 2}$$

$$= (AB)^{-1}$$

First we find AB.

Take R.H.S

$$B^{-1}A^{-1}$$

$$\Rightarrow AB = \begin{bmatrix} 11 & 10 \\ 7 & 5 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}_{2 \times 2}$$

$$\Rightarrow AB = \begin{bmatrix} (11)(1) + (10)(2) & (11)(3) + (10)(1) \\ (7)(1) + (5)(2) & (7)(3) + (5)(1) \end{bmatrix} = B = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}_{2 \times 2}$$

43

$$|B| = \begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix}$$

$$\text{Adj. of } A^{-1} = \frac{1}{|A|} \times \text{adj. of } A$$

$$|B| = (1)(1) - (2)(3)$$

$$A^{-1} = \frac{1}{-5} \begin{bmatrix} 5 & -10 \\ -7 & 11 \end{bmatrix}_{2 \times 2}$$

$$|B| = -5, |B| \neq 0 \text{ so it's inverse exist}$$

$$A^{-1} = \begin{bmatrix} 5 \times -1/5 & -10 \times -1/5 \\ -7 \times -1/5 & 11 \times -1/5 \end{bmatrix}_{2 \times 2}$$

$$\text{Adj. of } B = \begin{bmatrix} 1 & -3 \\ -2 & 1 \end{bmatrix}_{2 \times 2}$$

$$A^{-1} = \begin{bmatrix} -1/5 & 2/5 \\ 7/5 & -11/5 \end{bmatrix}_{2 \times 2}$$

$$B^{-1} = \frac{1}{|B|} \times \text{Adj. of } B$$

Now,

$$B^{-1} = \frac{1}{-5} \begin{bmatrix} 1 & -3 \\ -2 & 1 \end{bmatrix}_{2 \times 2}$$

$$= B^{-1}A^{-1} = \begin{bmatrix} -1/5 & 3/5 \\ 2/5 & -1/5 \end{bmatrix} \begin{bmatrix} -1/5 & 2/5 \\ 7/5 & -11/5 \end{bmatrix}$$

$$B^{-1} = \begin{bmatrix} -1/5 & 3/5 \\ 2/5 & -1/5 \end{bmatrix}$$

$$= \begin{bmatrix} (-1/5)(-1/5) + (3/5)(7/5) & (-1/5)(2/5) + (3/5)(-11/5) \\ (2/5)(-1/5) + (-1/5)(7/5) & (2/5)(2/5) + (-1/5)(-11/5) \end{bmatrix}_{2 \times 2}$$

$$A = \begin{bmatrix} 11 & 10 \\ 7 & 5 \end{bmatrix}_{2 \times 2}$$

$$|A| = \begin{vmatrix} 11 & 10 \\ 7 & 5 \end{vmatrix}$$

$$= \begin{bmatrix} 1/25 + 21/25 & -2/25 - 33/25 \\ -2/25 - 7/25 & 4/25 + 11/25 \end{bmatrix}_{2 \times 2}$$

$$|A| = (11)(5) - (7)(10)$$

$$= \begin{bmatrix} 26/25 & -43/25 \\ -17/25 & 31/25 \end{bmatrix}_{2 \times 2}$$

$$|A| = 55 - 70$$

$$|A| = -15, |A| \neq 0 \text{ so it's inverse exist.}$$

$$\text{Adj. of } A = \begin{bmatrix} 5 & -10 \\ -7 & 11 \end{bmatrix}_{2 \times 2}$$

44

$$(21) \text{ Adj } (AB) = (\text{adj } B) (\text{adj } A)$$

Take L.H.S

$$= \text{Adj } (AB)$$

$$= AB$$

$$= \begin{bmatrix} 11 & 10 \\ 7 & 5 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}_{2 \times 2}$$

$$= \begin{bmatrix} (11)(1) + (10)(2) & (11)(3) + (10)(1) \\ (7)(1) + (5)(2) & (7)(3) + (5)(1) \end{bmatrix}_{2 \times 2}$$

$$= \begin{bmatrix} 11+20 & 33+10 \\ 7+10 & 21+5 \end{bmatrix}_{2 \times 2}$$

$$= \begin{bmatrix} 31 & 43 \\ 17 & 26 \end{bmatrix}_{2 \times 2}$$

$$\text{Adj of } AB = \begin{bmatrix} 26 & -43 \\ -17 & 31 \end{bmatrix}_{2 \times 2}$$

Take R.H.S

$$= (\text{adj } B) (\text{adj } A)$$

$$\Rightarrow B = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}_{2 \times 2}$$

$$\Rightarrow \text{Adj of } B = \begin{bmatrix} 1 & -3 \\ -2 & 1 \end{bmatrix}_{2 \times 2}$$

$$A = \begin{bmatrix} 11 & 10 \\ 7 & 5 \end{bmatrix}_{2 \times 2}$$

$$\text{Adj of } A = \begin{bmatrix} 5 & -10 \\ -7 & 11 \end{bmatrix}_{2 \times 2}$$

Now

$$= (\text{adj } B) (\text{adj } A)$$

$$= \begin{bmatrix} 1 & -3 \\ -2 & 1 \end{bmatrix}_{2 \times 2} \begin{bmatrix} 5 & -10 \\ -7 & 11 \end{bmatrix}_{2 \times 2}$$

$$= \begin{bmatrix} (1)(5) + (-3)(-7) & (1)(-10) + (-3)(11) \\ (-2)(5) + (1)(-7) & (-2)(-10) + (1)(11) \end{bmatrix}_{2 \times 2}$$

$$= \begin{bmatrix} 5+21 & -10-33 \\ -10-7 & 20+11 \end{bmatrix}_{2 \times 2}$$

$$= \begin{bmatrix} 26 & -43 \\ -17 & 31 \end{bmatrix}_{2 \times 2}$$

Hence,

L.H.S = R.H.S proved!

(2) Verify the following -

$$(i) \left[\begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 0 \end{bmatrix} \right]^{-1} = \left[\begin{bmatrix} 3 & 1 \\ 2 & 0 \end{bmatrix} \right]^{-1} \left[\begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix} \right]^{-1}$$

Take L.H.S

$$= \begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix}_{2 \times 2} \begin{bmatrix} 3 & 1 \\ 2 & 0 \end{bmatrix}_{2 \times 2}$$

$$= \begin{bmatrix} (2)(3) + (3)(2) & (2)(1) + (3)(0) \\ (1)(3) + (-1)(2) & (1)(0) + (-1)(0) \end{bmatrix}_{2 \times 2}$$

$$\text{Adj of } A = \begin{bmatrix} 1 & -2 \\ -1 & 12 \end{bmatrix}_{2 \times 2}$$

$$A^{-1} = \frac{1}{|A|} \times \text{adj of } A$$

$$A^{-1} = \frac{1}{10} \begin{bmatrix} 1 & -2 \\ -1 & 12 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 1/10 & -1/5 \\ -1/10 & 6/5 \end{bmatrix}$$

45

$$= \begin{bmatrix} 6+6 & 2+0 \\ 3-2 & 1-0 \end{bmatrix}_{2 \times 2}$$

$$= \begin{bmatrix} 12 & 2 \\ 1 & 1 \end{bmatrix}_{2 \times 2}$$

Let,

$$\Rightarrow \begin{bmatrix} 12 & 2 \\ 1 & 1 \end{bmatrix}_{2 \times 2} = A$$

$$\Rightarrow |A| = \begin{vmatrix} 12 & 2 \\ 1 & 1 \end{vmatrix}$$

$$|A| = 12 - 2$$

$$\Rightarrow |A| = 10$$

Take R.H.S

$$\text{Let, } A = \begin{bmatrix} 2 & 1 \\ 2 & 0 \end{bmatrix}_{2 \times 2}$$

$$B = \begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix}_{2 \times 2}$$

For A^{-1} :

$$|A| = \begin{vmatrix} 2 & 1 \\ 2 & 0 \end{vmatrix}$$

$$|A| = (2)(0) - (2)(1)$$

$$|A| = -2$$

$$\text{Adj of } A = \begin{bmatrix} 0 & -1 \\ -2 & 3 \end{bmatrix}_{2 \times 2}$$

$$\Rightarrow A^{-1} = \frac{1}{|A|} \times \text{adj of } A$$

$$\Rightarrow A^{-1} = \frac{1}{-2} \begin{bmatrix} 0 & -1 \\ -2 & 3 \end{bmatrix}_{2 \times 2}$$

$$\Rightarrow A^{-1} = \begin{bmatrix} 0 & 1/2 \\ 1 & -3/2 \end{bmatrix}_{2 \times 2}$$

For B^{-1} :

$$|B| = \begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix}$$

$$|B| = (2)(-1) - (3)(1)$$

$$|B| = -5$$

$$\text{Adj of } B = \begin{bmatrix} -1 & -3 \\ -1 & 2 \end{bmatrix}_{2 \times 2}$$

$$B^{-1} = \frac{1}{|B|} \times \text{adj of } B$$

$$B^{-1} = \frac{1}{-5} \begin{bmatrix} -1 & -3 \\ -1 & 2 \end{bmatrix}_{2 \times 2}$$

$$B^{-1} = \begin{bmatrix} 1/5 & 3/5 \\ 1/5 & -2/5 \end{bmatrix}_{2 \times 2}$$

$$A^{-1}B^{-1} = \begin{bmatrix} 0 & 1/2 \\ -2 & -3/2 \end{bmatrix}_{2 \times 2} \begin{bmatrix} 1/5 & 3/5 \\ 1/5 & -2/5 \end{bmatrix}_{2 \times 2}$$

$$A^{-1}B^{-1} = \begin{bmatrix} (0)(1/5) + (1/2)(1/5) & (0)(3/5) + (1/2)(-2/5) \\ (-2)(1/5) + (-3/2)(1/5) & (-2)(3/5) + (-3/2)(-2/5) \end{bmatrix}_{2 \times 2}$$

(46)

$$A^{-1}B^{-1} = \begin{bmatrix} 10+30 & 30-60 \\ 10-20 & 30+40 \end{bmatrix}_{2 \times 2}$$

$$A^{-1}B^{-1} = \begin{bmatrix} 40 & -30 \\ -10 & 70 \end{bmatrix}_{2 \times 2}$$

$$A^{-1}B^{-1} = \begin{bmatrix} 0 & 1/2 \\ 1 & -3/2 \end{bmatrix}_{2 \times 2} \begin{bmatrix} 1/5 & 3/5 \\ 1/5 & -2/5 \end{bmatrix}_{2 \times 2}$$

$$A^{-1}B^{-1} = \begin{bmatrix} (0)(1/5) + (1/2)(1/5) & (0)(3/5) + (1/2)(-2/5) \\ (1)(1/5) + (-3/2)(1/5) & (1)(3/5) + (-3/2)(-2/5) \end{bmatrix}_{2 \times 2}$$

$$A^{-1}B^{-1} = \begin{bmatrix} 0+1/10 & 0-2/10 \\ 1/5-3/10 & 3/5+6/10 \end{bmatrix}_{2 \times 2}$$

$$A^{-1}B^{-1} = \begin{bmatrix} 1/10 & -1/5 \\ -1/10 & 6/5 \end{bmatrix}_{2 \times 2}$$

Hence,

L.H.S. = R.H.S. proved!

$$(x) \left[\begin{bmatrix} 1 & 3 \\ i & 1 \end{bmatrix} \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix} \right]^{-1} = \left[\begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix} \right]^{-1} \left[\begin{bmatrix} 1 & 3 \\ i & 1 \end{bmatrix} \right]^{-1}$$

So

$$\text{Take L.H.S.} = \begin{bmatrix} 1+1 & -i+i \\ i-i & 1+1 \end{bmatrix}_{2 \times 2}$$

$$= \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}_{2 \times 2}$$

$$= \begin{bmatrix} 1 & 3 \\ i & 1 \end{bmatrix} \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix}$$

$$= \begin{bmatrix} (1)(1)+(3)(-i) & (1)(-i)+(3)(1) \\ (i)(1)+(1)(-i) & (i)(-i)+(1)(1) \end{bmatrix}_{2 \times 2} \quad \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = A$$

(47)

$$|A| = \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix}$$

$$|A| = 4 - 0 = 4 \neq 0, \text{ so } A^{-1} \text{ exists.}$$

$$\text{adj of } A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}_{2 \times 2}$$

$$A^{-1} = \frac{1}{|A|} \times \text{adj of } A$$

$$A^{-1} = \frac{1}{4} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}_{2 \times 2}$$

$$A^{-1} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}_{2 \times 2}$$

Take R.H.S.

$$= \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 3 \\ i & 1 \end{bmatrix}^{-1}$$

let,

$$A = \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix}_{2 \times 2}$$

$$B = \begin{bmatrix} 1 & 3 \\ i & 1 \end{bmatrix}_{2 \times 2}$$

$$|A| = \begin{vmatrix} 1 & -i \\ -i & 1 \end{vmatrix}$$

$$|A| = 1+1=2, |A| \neq 0, \text{ so } A^{-1} \text{ exists.}$$

$$\text{adj of } A = \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix}_{2 \times 2}$$

$$A^{-1} = \frac{1}{|A|} \times \text{adj of } A$$

$$A^{-1} = \frac{1}{2} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix}_{2 \times 2}$$

$$A^{-1} = \begin{bmatrix} 1/2 & i/2 \\ i/2 & 1/2 \end{bmatrix}_{2 \times 2}$$

For B⁻¹:

$$|B| = \begin{vmatrix} 1 & 3 \\ i & 1 \end{vmatrix}$$

$$|B| = 1+1=2, |B| \neq 0, \text{ so } B^{-1} \text{ exists.}$$

$$\text{adj of } B = \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix}_{2 \times 2}$$

$$B^{-1} = \frac{1}{|B|} \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix}_{2 \times 2}$$

$$B^{-1} = \begin{bmatrix} 1/2 & -i/2 \\ -i/2 & 1/2 \end{bmatrix}_{2 \times 2}$$

48

$$A^{-1}B^{-1} = \begin{bmatrix} 1/2 & 3/2 \\ 3/2 & 1/2 \end{bmatrix}_{2 \times 2} \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix}_{2 \times 2}$$

$$A^{-1}B^{-1} = \begin{bmatrix} (1/2)(1/2) + (3/2)(-1/2) & (1/2)(-1/2) + (3/2)(1/2) \\ (3/2)(1/2) + (1/2)(-1/2) & (3/2)(-1/2) + (1/2)(1/2) \end{bmatrix}_{2 \times 2}$$

$$A^{-1}B^{-1} = \begin{bmatrix} 1/4 + 3/4 & -1/4 + 3/4 \\ 3/4 - 1/4 & 3/4 + 1/4 \end{bmatrix}_{2 \times 2}$$

$$A^{-1}B^{-1} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}_{2 \times 2} \text{ e.g.}$$

Hence, Proved!

⑦ Calculate the inverse by using adjoint method.

$$(i) \begin{bmatrix} 5 & 6 \\ 3 & 4 \end{bmatrix}$$

Sol

$$\text{let, } A = \begin{bmatrix} 5 & 6 \\ 3 & 4 \end{bmatrix}_{2 \times 2}$$

$$\therefore A^{-1} = \frac{1}{|A|} \times \text{adj of } A$$

$$|A| = \begin{vmatrix} 5 & 6 \\ 3 & 4 \end{vmatrix}$$

$$|A| = 20 - 18$$

$$|A| = 2 \quad |A| \neq 0, \text{ So it's inverse exist}$$

$$\text{adj of } A = \begin{bmatrix} 4 & -6 \\ -3 & 5 \end{bmatrix}_{2 \times 2}$$

$$A^{-1} = \frac{1}{2} \begin{bmatrix} 4 & -6 \\ -3 & 5 \end{bmatrix}_{2 \times 2}$$

$$A^{-1} = \begin{bmatrix} 2 & -3 \\ -3/2 & 5/2 \end{bmatrix}_{2 \times 2} \text{ e.g.}$$

$$(ii) \begin{bmatrix} 7 & 3 \\ 9 & 6 \end{bmatrix}$$

Sol

let,

$$A = \begin{bmatrix} 7 & 3 \\ 9 & 6 \end{bmatrix}_{2 \times 2}$$

$$\therefore A^{-1} = \frac{1}{|A|} \times \text{adj of } A$$

$$|A| = \begin{vmatrix} 7 & 3 \\ 9 & 6 \end{vmatrix}$$

49

$$|A| = 42 - 27$$

$$|A| = 15$$

$$\text{adj of } A = \begin{bmatrix} 6 & -5 \\ -9 & 7 \end{bmatrix}$$

$$A^{-1} = \frac{1}{15} \begin{bmatrix} 6 & -5 \\ -9 & 7 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 2/5 & -1/3 \\ -3/5 & 7/15 \end{bmatrix}_{2 \times 2}$$

$$(iii) \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 0 & 9 \end{bmatrix}$$

Sol

let,

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 0 & 9 \end{bmatrix}_{3 \times 3}$$

$$\therefore A^{-1} = \frac{1}{|A|} \times \text{adj of } A$$

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 0 & 9 \end{vmatrix}$$

$$|A| = 1(4 \cdot 0 - 2 \cdot 2 \cdot 6) + 3(2 \cdot 4 \cdot 3)$$

$$|A| = 1(3 \cdot 6 - 0) - 2(18 - 18) + 3(0 - 12)$$

$$|A| = 36 - 36$$

$$|A| = 0$$

$|A| = 0$, So it's inverse doesn't exist.

$$(iv) \begin{bmatrix} 2 & 3 & 4 \\ 3 & 1 & 5 \\ -5 & 1 & 0 \end{bmatrix}$$

Sol

let,

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 1 & 5 \\ -5 & 1 & 0 \end{bmatrix}_{3 \times 3}$$

$$\therefore A^{-1} = \frac{1}{|A|} \times \text{adj of } A$$

$$|A| = \begin{vmatrix} 2 & 3 & 4 \\ 3 & 1 & 5 \\ -5 & 1 & 0 \end{vmatrix}$$

$$|A| = 2(1 \cdot 0 - 3 \cdot 5) + 3(0 - 5) + 4(3 \cdot 1)$$

$$|A| = 2(0 - 15) - 3(0 + 25) + 4(3 + 5)$$

$$|A| = -10 - 75 + 28$$

$$|A| = -57 \quad \text{As, } |A| \neq 0, \text{ So it's inverse exist}$$

$$\text{adj of } A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}^t$$

Co-factors of A:

$$\therefore A_{ij} = (-1)^{i+j} M_{ij}$$

(50)

$$A_{11} = (-1)^{1+1} \begin{vmatrix} 1 & 5 \\ 1 & 0 \end{vmatrix}$$

$$A_{11} = (-1)^1 (0-5)$$

$$A_{11} = -5$$

$$A_{12} = (-1)^{1+2} \begin{vmatrix} 3 & 5 \\ -5 & 0 \end{vmatrix}$$

$$A_{12} = (-1)^3 (0+25)$$

$$A_{12} = -25$$

$$A_{13} = (-1)^{1+3} \begin{vmatrix} 3 & 1 \\ -5 & 1 \end{vmatrix}$$

$$A_{13} = (-1)^4 (3+5)$$

$$A_{13} = 8$$

$$A_{21} = (-1)^{2+1} \begin{vmatrix} 3 & 4 \\ 1 & 0 \end{vmatrix}$$

$$A_{21} = (-1)^3 (0-4)$$

$$A_{21} = 4$$

$$A_{22} = (-1)^{2+2} \begin{vmatrix} 2 & 4 \\ -5 & 0 \end{vmatrix}$$

$$A_{22} = (-1)^4 (0+20)$$

$$A_{22} = 20$$

$$A_{23} = (-1)^{2+3} \begin{vmatrix} 2 & 3 \\ -5 & 1 \end{vmatrix}$$

$$A_{23} = (-1)^5 (2+15)$$

$$A_{23} = -17$$

$$A_{31} = (-1)^{3+1} \begin{vmatrix} 3 & 4 \\ 1 & 5 \end{vmatrix}$$

$$A_{31} = (-1)^4 (15-4)$$

$$A_{31} = 11$$

$$A_{32} = (-1)^{3+2} \begin{vmatrix} 2 & 4 \\ 3 & 5 \end{vmatrix}$$

$$A_{32} = (-1)^{5} (10-12)$$

$$A_{32} = 2$$

$$A_{33} = (-1)^{3+3} \begin{vmatrix} 2 & 3 \\ 3 & 1 \end{vmatrix}$$

$$A_{33} = (-1)^6 (2-9)$$

$$A_{33} = -7$$

$$\text{adj of } A = \begin{bmatrix} -5 & -25 & 8 \\ 4 & 20 & -17 \\ 11 & 2 & -7 \end{bmatrix}^T$$

$$\text{adj of } A = \begin{bmatrix} -5 & 4 & 11 \\ -25 & 20 & 2 \\ 8 & -17 & -7 \end{bmatrix}^{3 \times 3}$$

$$A^{-1} = \frac{1}{|A|} \begin{bmatrix} -5 & 4 & 11 \\ -25 & 20 & 2 \\ 8 & -17 & -7 \end{bmatrix}^{3 \times 3}$$

$$A^{-1} = \frac{1}{-5} \begin{bmatrix} 5/52 & -4/52 & -11/52 \\ 25/52 & -20/52 & -2/52 \\ -8/52 & 17/52 & 7/52 \end{bmatrix}^{3 \times 3}$$

$$M = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 1 & -1 & 1 \end{bmatrix}$$

$$\text{let } A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 1 & -1 & 1 \end{bmatrix}^{3 \times 3}$$

$$A^{-1} = \frac{\text{adj of } A}{|A|}$$

$$|A| = \begin{vmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 1 & -1 & 1 \end{vmatrix}$$

$$|A| = 1 \begin{vmatrix} 2 & 1 & -0 \\ -1 & 1 & 1 \end{vmatrix} + 2 \begin{vmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} + 2 \begin{vmatrix} 0 & 2 & 1 \\ 1 & 1 & 1 \end{vmatrix}$$

$$|A| = 1(2+1) - 0(0-1) + 2(0-1)$$

$$|A| = 2 - 0 - 2$$

$$|A| = 1 \quad \text{As } |A| \neq 0, \text{ so it's}$$

$$\text{inverse exist. } A^{-1} = (-1)^3 (0+1)$$

$$A^{-1} = -2$$

(51)

$$\text{adj of } A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^T$$

$$\text{Co-factor of } A_i$$

$$\therefore A_{ij} = (-1)^{i+j} M_{ij}$$

$$A_{11} = (-1)^{1+1} \begin{vmatrix} 2 & 1 \\ -1 & 1 \end{vmatrix}$$

$$A_{11} = (-1)^2 (2+1)$$

$$A_{11} = 3$$

$$A_{12} = (-1)^{1+2} \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix}$$

$$A_{12} = (-1)^3 (0-1)$$

$$A_{12} = 1$$

$$A_{13} = (-1)^{1+3} \begin{vmatrix} 0 & 2 \\ 1 & -1 \end{vmatrix}$$

$$A_{13} = (-1)^4 (0-2)$$

$$A_{13} = -2$$

$$A_{21} = (-1)^{2+1} \begin{vmatrix} 0 & 2 \\ -1 & 1 \end{vmatrix}$$

$$A_{21} = (-1)^3 (0+1)$$

$$A_{21} = -2$$

(S2)

$$A_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix}$$

$$A_{22} = (-1)^4 (1-2)$$

$$A_{22} = -1$$

$$A_{23} = (-1)^{2+3} \begin{vmatrix} 1 & 0 \\ 1 & -1 \end{vmatrix}$$

$$A_{23} = (-1)^5 (-1-0)$$

$$A_{23} = 1$$

$$A_{31} = (-1)^{3+1} \begin{vmatrix} 0 & 2 \\ 2 & 1 \end{vmatrix}$$

$$A_{31} = (-1)^4 (0-4)$$

$$A_{31} = -4$$

$$A_{32} = (-1)^{3+2} \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix}$$

$$A_{32} = (-1)^5 (1-0)$$

$$A_{32} = -1$$

$$A_{33} = (-1)^{3+3} \begin{vmatrix} 1 & 0 \\ 0 & 2 \end{vmatrix}$$

$$A_{33} = (-1)^6 (2-0)$$

$$A_{33} = 2$$

$$\text{adj of } A = \begin{bmatrix} 3 & 1 & -2 \\ -2 & -1 & 1 \\ -4 & -1 & 2 \end{bmatrix}^{t}_{3 \times 3}$$

$$\text{adj of } A = \begin{bmatrix} 3 & -2 & -4 \\ 1 & -1 & -1 \\ -2 & 1 & 2 \end{bmatrix}_{3 \times 3}$$

$$A^{-1} = \frac{1}{1} \begin{bmatrix} 3 & -2 & -4 \\ 1 & -1 & -1 \\ -2 & 1 & 2 \end{bmatrix}_{3 \times 3}$$

$$A^{-1} = \begin{bmatrix} 3 & -2 & -4 \\ 1 & -1 & -1 \\ -2 & 1 & 2 \end{bmatrix}_{3 \times 3}$$

$\frac{1}{2}$

$$\text{Min } \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{bmatrix}$$

S-1

$$\text{let, } A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{bmatrix}_{3 \times 3}$$

$$A^{-1} = \frac{\text{adj of } A}{|A|}$$

|A|

$$|A| = \begin{vmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{vmatrix}$$

$$|A| = 2 \begin{vmatrix} 1 & 0 \\ -3 & 5 \end{vmatrix} - 1 \begin{vmatrix} 1 & 0 \\ 2 & 5 \end{vmatrix} + 0 \begin{vmatrix} 1 & 1 \\ 2 & -3 \end{vmatrix}$$

(S3)

$$|A| = 2(5+0) - 1(5-0) + 0(-3-2)$$

$$|A| = 10 - 5$$

$$|A| = 5$$

As, $|A| \neq 0$. So it's

Inverse exist

$$\text{adj of } A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^t$$

Co-factors of A:

$$A_{ij} = (-1)^{i+j} M_{ij}$$

$$A_{11} = (-1)^{1+1} \begin{vmatrix} 1 & 0 \\ -3 & 5 \end{vmatrix}$$

$$A_{11} = (-1)^2 (5+0)$$

$$A_{11} = 5$$

$$A_{12} = (-1)^{1+2} \begin{vmatrix} 1 & 0 \\ 2 & 5 \end{vmatrix}$$

$$A_{12} = (-1)^3 (5+0)$$

$$A_{12} = -5$$

$$A_{13} = (-1)^{1+3} \begin{vmatrix} 1 & 1 \\ 2 & -3 \end{vmatrix}$$

$$A_{13} = (-1)^4 (-3-2)$$

$$A_{13} = -5$$

$$A_{21} = (-1)^{2+1} \begin{vmatrix} 1 & 0 \\ -3 & 5 \end{vmatrix}$$

$$A_{21} = (-1)^3 (5+0)$$

$$A_{21} = -5$$

$$A_{22} = (-1)^{2+2} \begin{vmatrix} 2 & 0 \\ 2 & -3 \end{vmatrix}$$

$$A_{22} = (-1)^4 (10-0)$$

$$A_{22} = 10$$

$$A_{23} = (-1)^{2+3} \begin{vmatrix} 2 & 1 \\ 2 & -3 \end{vmatrix}$$

$$A_{23} = (-1)^5 (-6-2)$$

$$A_{23} = 8$$

$$A_{31} = (-1)^{3+1} \begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix}$$

$$A_{31} = (-1)^4 (0-0)$$

$$A_{31} = 0$$

$$A_{32} = (-1)^{3+2} \begin{vmatrix} 2 & 0 \\ 1 & 0 \end{vmatrix}$$

$$A_{32} = (-1)^5 (0-0)$$

$$A_{32} = 0$$

Your Success is our Win (54)

$$A_{33} = (-1)^{2+3} \begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix}$$

$$A_{33} = (-1)^4 (2-1)$$

$$A_{33} = 1$$

$$\text{adj of } A = \begin{bmatrix} 5 & -5 & -5 \\ -5 & 10 & 8 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3}$$

$$\text{adj of } A = \begin{bmatrix} 5 & 5 & 0 \\ -5 & 10 & 0 \\ -5 & 8 & 1 \end{bmatrix}_{3 \times 3}$$

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 5 & -5 & 0 \\ -5 & 10 & 0 \\ -5 & 8 & 1 \end{bmatrix}_{3 \times 3}$$

$$A^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & 0 \\ -1 & 8/5 & 1/5 \end{bmatrix}_{3 \times 3}$$

Ex # 2.4

Q) Let $A = \begin{bmatrix} 2 & 5 & 0 \\ 3 & 4 & 6 \\ 1 & -5 & 1 \end{bmatrix}$, verify that $|A| = |A^T|$

Sol

Take L.H.S

$$= |A|$$

$$\Rightarrow A = \begin{bmatrix} 2 & 5 & 0 \\ 3 & 4 & 6 \\ 1 & -5 & 1 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow |A| = \begin{vmatrix} 2 & 5 & 0 \\ 3 & 4 & 6 \\ 1 & -5 & 1 \end{vmatrix}$$

$$\Rightarrow |A| = 2 \begin{vmatrix} 4 & 6 & -5 \\ 3 & 1 & 1 \end{vmatrix} - 5 \begin{vmatrix} 3 & 6 & 0 \\ 1 & 1 & -5 \end{vmatrix} + 0 \begin{vmatrix} 3 & 4 \\ 1 & -5 \end{vmatrix}$$

$$\Rightarrow |A| = 2(4+30) - 5(3-6) + 0(-15-4)$$

$$\Rightarrow |A| = 2(34) - 5(-3) + 0$$

$$\Rightarrow |A| = 68 + 15$$

$$\Rightarrow |A| = 83$$

Take R.H.S

$$= |A^T|$$

$$\Rightarrow A^T = \begin{bmatrix} 2 & 3 & 1 \\ 5 & 4 & 6 \\ 0 & -5 & 1 \end{bmatrix}_{3 \times 3}$$

(55)

$$\Rightarrow A^T = \begin{bmatrix} 2 & 3 & 1 \\ 5 & 4 & -5 \\ 0 & 6 & 1 \end{bmatrix}_{3 \times 3}$$

$$|A| = 2(4+30) - 5(3) + 1(30)$$

$$|A| = 2(34) - 15 + 30$$

$$|A| = 67 + 15$$

$$|A| = 83$$

$$L.H.S = R.H.S$$

$$\Rightarrow |A^T| = \begin{vmatrix} 2 & 3 & 1 \\ 5 & 4 & -5 \\ 0 & 6 & 1 \end{vmatrix}$$

$$\Rightarrow |A^T| = 2 \begin{vmatrix} 4 & -5 \\ 6 & 1 \end{vmatrix} - 3 \begin{vmatrix} 5 & -5 \\ 0 & 1 \end{vmatrix} + 1 \begin{vmatrix} 5 & 4 \\ 6 & 6 \end{vmatrix}$$

Hence, proved!

Q) Without expanding, prove each of the following.

(i) $\begin{vmatrix} x+y & y+z & z+x \\ 2 & x & y \\ 1 & 1 & 1 \end{vmatrix} = 0$

Sol

By adding R_2 into R_1 , we get

$$\Rightarrow \begin{vmatrix} x+y+z & x+y+z & x+y+z \\ 2 & x & y \\ 1 & 1 & 1 \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} x+y+z & 1 & 1 & 1 \\ 2 & x & y \\ 1 & 1 & 1 \end{vmatrix}$$

$\therefore$ two rows are identical

$\therefore$ Determinant = 0

$$\Rightarrow x+y+z(0) = 0$$

$$\Rightarrow 0 = 0$$

Hence, proved!